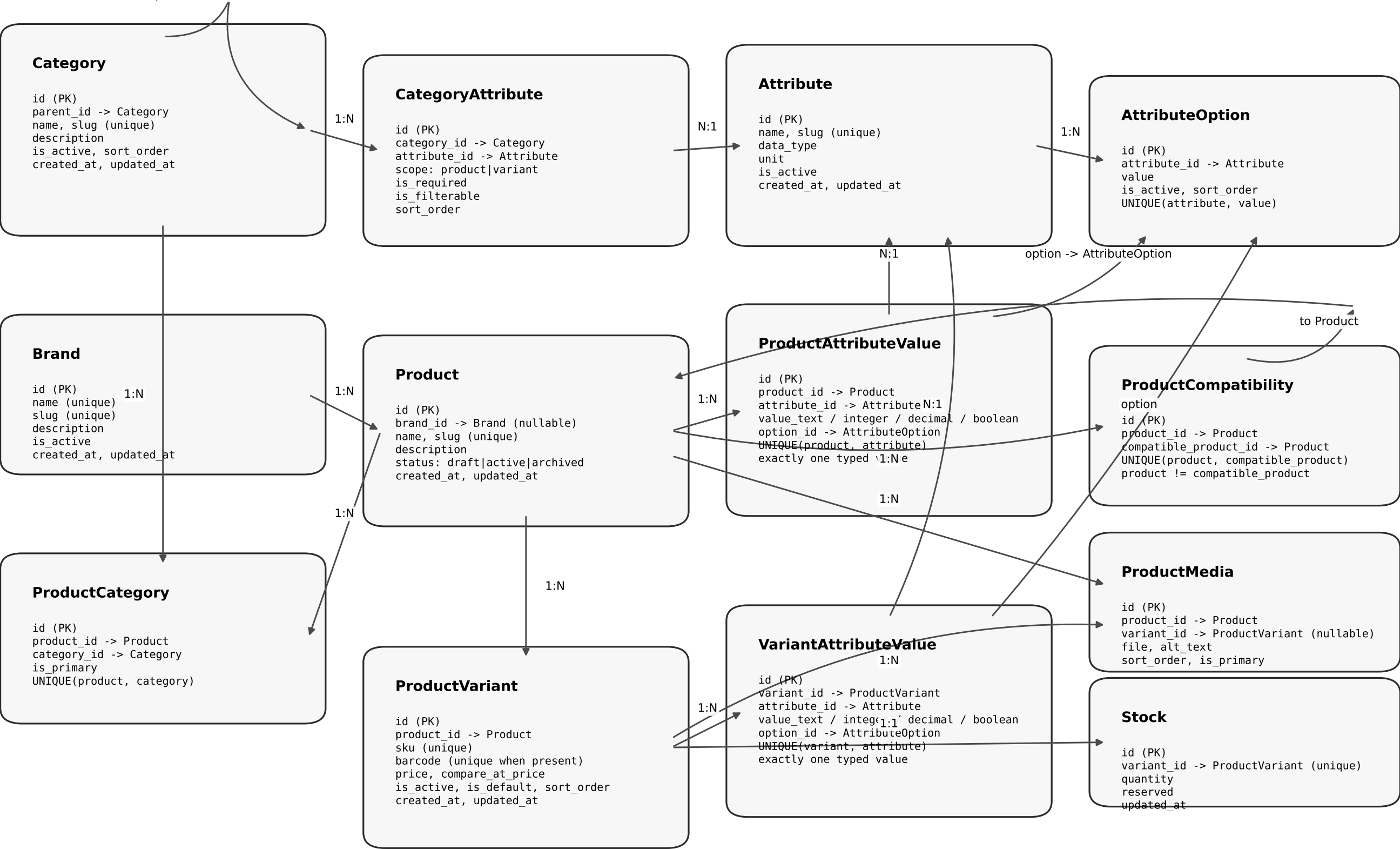


Phone Accessories — Catalog DB Schema

Source: CATALOG_SCHEMA.md (approved target design)



Legend: Product = conceptual product/model; ProductVariant = concrete sellable SKU.
Price and stock live on ProductVariant. Main characteristics use the Attribute system.
ProductMedia stores storage-backed image references, not binaries in PostgreSQL.

Schema notes and key invariants

Condensed from CATALOG_SCHEMA.md

Core ideas

- Use one universal Product model. Do not create separate Phone/Laptop/Accessory tables.
- Product is the model. ProductVariant is the concrete SKU.
- Price and stock belong to ProductVariant, not Product.
- Characteristics are extensible via Attribute + CategoryAttribute + typed value tables.
- Do not store main filterable specs in fixed columns or one JSON field.

Important constraints

- Category.slug and Brand.slug are unique.
- Category.parent uses PROTECT.
- ProductCategory.category uses PROTECT.
- One primary category per product (conditional unique).
- One default variant per product (conditional unique).
- Stock is one row per variant; available = quantity - reserved.
- ProductCompatibility is directed and cannot point to self.

Application-level invariants

- ProductMedia.variant.product_id must equal ProductMedia.product_id.
- Typed value field must match Attribute.data_type.
- AttributeOption.attribute_id must match the Attribute on the value row.
- Publication workflow may later require category/default variant/required attributes.

Not in current scope

- Cart / checkout / orders
- Payments / shipments / addresses
- Discounts / coupons
- Multiple currencies
- Multiple warehouses / stock ledger
- Reviews / ratings
- SEO subsystem
- Media processing pipeline
- Production S3 configuration

Recommended implementation order

- 1. Category + Brand
- 2. Product + ProductCategory + ProductVariant
- 3. Attribute + AttributeOption + CategoryAttribute
- 4. ProductAttributeValue + VariantAttributeValue
- 5. ProductMedia
- 6. Stock
- 7. ProductCompatibility

Practical note for the team

- This PDF is a visual aid only.
- The source of truth remains CATALOG_SCHEMA.md.
- When schema and code diverge, compare repository code + migrations against CATALOG_SCHEMA.md.